EEG Brain Speller Project Report

-By Sarthak Gupta (240939)

Project Overview

The following project implements an EEG Based Brain Speller, complete with speller ui functionality. It enables a person to type characters using only brain signals.

Dataset Used

While the requirements docs says to use MOABB, I couldn't find viable datasets inside it which provided 'TargetChar' labels or the character sequences needed for end-to-end decoding.

Hence I used a dataset which is well known and one which we have used in an earlier assignment during the project as well, being the **BCI Competition III Dataset II**, since it has the full structure, StimulusCode, Flashing, TargetChar, and the raw signal all aligned properly.

Step by Step Implementation

Dataloader.py:

We have two subjects in dataset II, being Subject A and Subject B, each with their own train and test files.

The dataloder reads the .mat files and extracts Signal, StimulusCode, Flashing, StimulusType(train only) and TargetChar(train only). The StimulusType field offers the ground-truth binary P300 labels as provided by the dataset. This is significantly more robust than attempting to find them manually using StimulusCode.

Preprocess.py:

Since in the chosen dataset the signal is already structured (as n_chars,n_samples,n_channels), no MNE conversion or re-epoching was needed. The preprocess file handles epoch extraction directly from the numpy array.

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Before epoch extraction following filters are used:
Butterworth Bandpass filter: (0.1-30hz) preserves the P300 component while removing the higher freq noise, since the P300 signal sits mostly between 0.1-10hz.
Notch Filter: 50hz, removes the powerline interference which is the standard in India where mains freq is 50hz.

For each character trial, flash onsets are detected as 0->1 transitions in the flashing array. At each onset, a 240hz 1sec sample window is extracted which gives 180 epochs per character (if using 15 repetitions). Epochs that would skip the trial boundary are skipped.

Features.py:

Xdawn spatial filtering and downsampling both have been implemented here.

For LDA and SVM, each epoch is windowed to samples 20-120 , corresponding to roughly 83-500ms post-stimulus. The window starts at sample 20 rather than 0 to exclude the pre stimulus baseline noise, and ends at 120 to focus on the P300 latency range. Going beyond 500ms adds noise rather than useful signal. Then it is downsampled from 100 to 25 time points per channel, which makes it quicker for LDA.

Each windowed epoch is transposed to (n_channels, n_times) and passed through Xdawn spatial filtering. First it is fit on the full training set with the ground truth labels, then it is applied to both train and test data. 6 components were retained as a balance between capturing the dominant P300 signal and discarding noise. The filtered output is then flattened to a 1D feature vector per epoch and fed to the classifier.

For EEGNet, Xdawn is intentionally skipped since this model learns its own spatial filters internally via depthwise convolutions. A window of samples

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20-180 is used here. The 3D shape (n_trials, n_channels, n_times) is preserved as EEGNet requires it.

Models.py:

Three models have been implemented, being LDA,SVM and EEGNET. These are switchable via parameter in main.py.

More discussion on accuracies and ITR of the various models is done in the analysis section of the report.

Evaluate.py:

Implements all metrics required by the doc. Epoch-level precision, recall, and F1 are computed on the training set. Due to the heavy class imbalance (~1 P300 per 11 non-P300 epochs), recall is expectedly lower than precision. Stratified 5-fold cross-validation preserves this imbalance across folds. A confusion matrix is saved to the results folder.

Character-level accuracy and ITR are computed on the test set. ITR uses the standard BCI formula with N=36 and T derived from repetition count. A rep sweep (3, 5, 7, 10, 15) is run to analyse the accuracy-ITR tradeoff.

Speller_ui.py:

The speller UI implements a visual P300-based character display using PsychoPy. A 6×6 grid of characters is shown on screen, along with a running decoded output string.

During each character prediction, rows and columns of the grid are flashed in random order for a fixed number of repetitions. Each flash briefly highlights a row or column, simulating the standard P300 speller paradigm. The timing used is 100 ms ON and 75 ms OFF per flash.

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After the flashing sequence for a character completes, the decoded character (obtained from the model) is appended to the output text and displayed at the top of the screen. This process repeats for the full predicted string.

The UI runs subject-wise, showing an intro screen before starting and a final decoded output screen at the end. It is integrated into the main pipeline and is optionally triggered after decoding using a flag in main.py.

Main.py:

This incorporates all the various files, loads data, runs preprocessing and feature extraction, trains the selected model and evaluates the testset. Model can be changed using the MODE variable.

Each subject is tested and trained independently.

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RESULTS

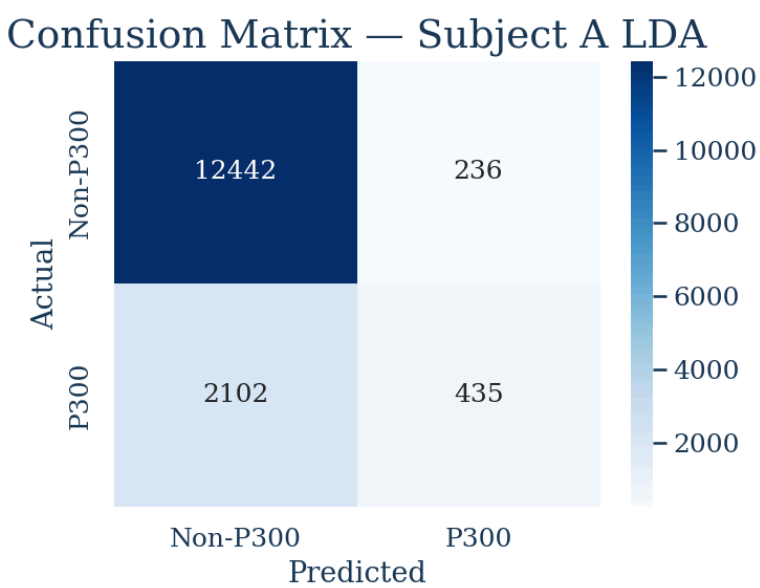
LDA:

Subject A:

```
--- Epoch-level Metrics ---
Precision : 0.6553
Recall    : 0.1514
F1 Score  : 0.2459


--- 5-Fold Stratified Cross-Validation ---
Accuracy  : 0.8383 ± 0.0029
Precision : 0.5646 ± 0.0406
Recall    : 0.1305 ± 0.0131
F1 Score  : 0.2118 ± 0.0195
Confusion matrix saved to results/confusion_matrix.png

--- Character Decoding ---
Character Accuracy: 35.00% (35/100)
  reps=3 | ITR       : 8.59 bits/minute  (N=36, T=6.3s, reps=3)
Character Accuracy: 54.00% (54/100)
  reps=5 | ITR       : 10.37 bits/minute  (N=36, T=10.5s, reps=5)
Character Accuracy: 65.00% (65/100)
  reps=7 | ITR       : 9.96 bits/minute  (N=36, T=14.7s, reps=7)
Character Accuracy: 76.00% (76/100)
  reps=10 | ITR       : 8.98 bits/minute  (N=36, T=21.0s, reps=10)
Character Accuracy: 91.00% (91/100)
  reps=15 | ITR       : 8.14 bits/minute  (N=36, T=31.5s, reps=15)
```



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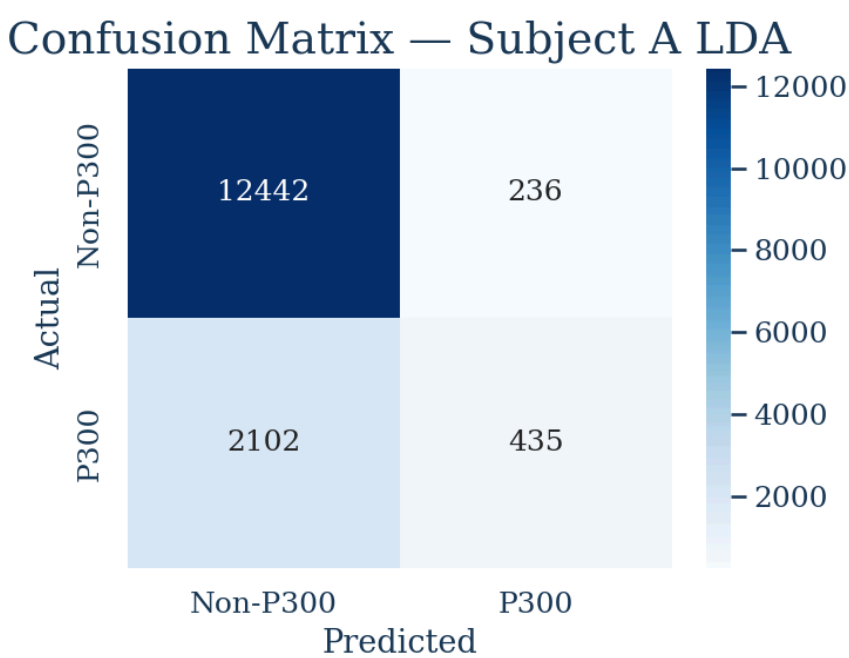
Subject B:

```
--- Epoch-level Metrics ---
Precision : 0.7332
Recall    : 0.4562
F1 Score  : 0.5625


--- 5-Fold Stratified Cross-Validation ---
Accuracy  : 0.8735 ± 0.0048
Precision : 0.6950 ± 0.0196
Recall    : 0.4294 ± 0.0235
F1 Score  : 0.5307 ± 0.0233
Confusion matrix saved to results/confusion_matrix.png

--- Character Decoding ---
Character Accuracy: 70.00% (70/100)
  reps=3 | ITR       : 26.19 bits/minute  (N=36, T=6.3s, reps=3)
Character Accuracy: 82.00% (82/100)
  reps=5 | ITR       : 20.38 bits/minute  (N=36, T=10.5s, reps=5)
Character Accuracy: 85.00% (85/100)
  reps=7 | ITR       : 15.47 bits/minute  (N=36, T=14.7s, reps=7)
Character Accuracy: 88.00% (88/100)
  reps=10 | ITR       : 11.50 bits/minute  (N=36, T=21.0s, reps=10)
Character Accuracy: 90.00% (90/100)
  reps=15 | ITR       : 7.98 bits/minute  (N=36, T=31.5s, reps=15)
```



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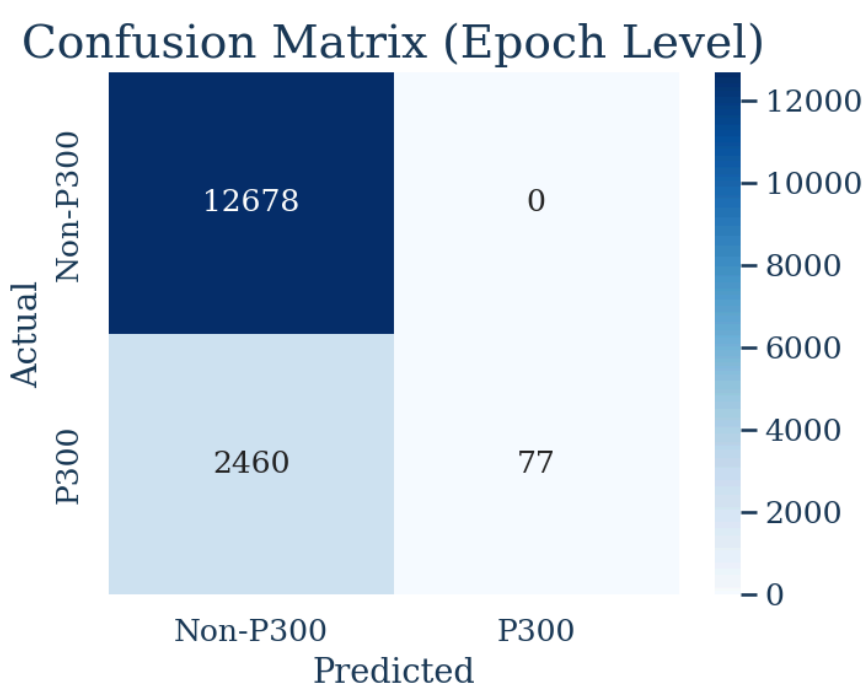
SVM:

Subject A:

```
--- Epoch-level Metrics ---
Precision : 1.0000
Recall    : 0.0304
F1 Score  : 0.0589


--- 5-Fold Stratified Cross-Validation ---
Accuracy  : 0.8337 ± 0.0005
Precision : 0.6000 ± 0.4899
Recall    : 0.0028 ± 0.0024
F1 Score  : 0.0055 ± 0.0047
Confusion matrix saved to results/confusion_matrix.png

--- Character Decoding ---
Character Accuracy: 37.00% (37/100)
  reps=3 | ITR       : 9.41 bits/minute  (N=36, T=6.3s, reps=3)
Character Accuracy: 45.00% (45/100)
  reps=5 | ITR       : 7.75 bits/minute  (N=36, T=10.5s, reps=5)
Character Accuracy: 62.00% (62/100)
  reps=7 | ITR       : 9.24 bits/minute  (N=36, T=14.7s, reps=7)
Character Accuracy: 74.00% (74/100)
  reps=10 | ITR       : 8.60 bits/minute  (N=36, T=21.0s, reps=10)
Character Accuracy: 84.00% (84/100)
  reps=15 | ITR       : 7.08 bits/minute  (N=36, T=31.5s, reps=15)
```



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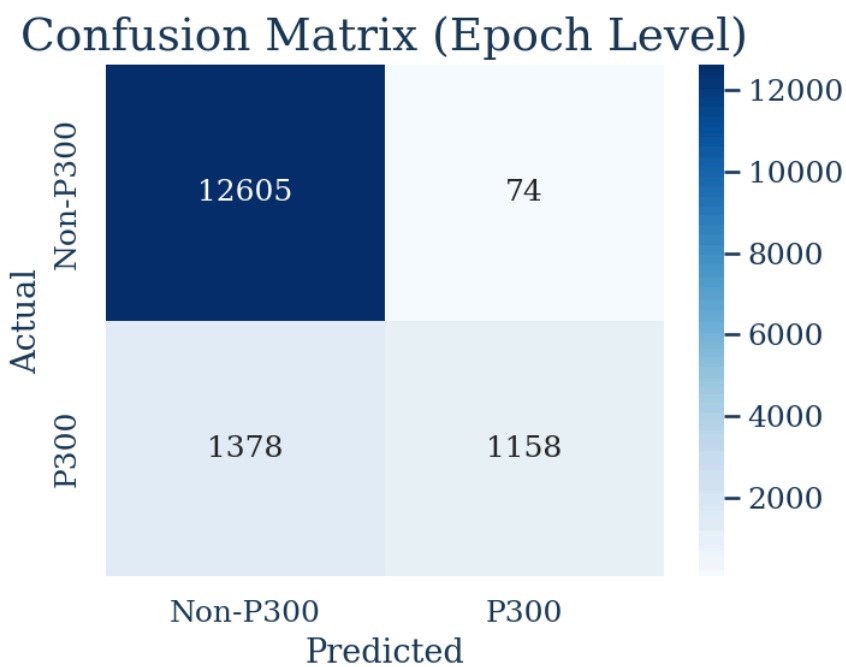
Subject B:

```
--- Epoch-level Metrics ---
Precision : 0.9399
Recall    : 0.4566
F1 Score  : 0.6146


--- 5-Fold Stratified Cross-Validation ---
Accuracy  : 0.8699 ± 0.0031
Precision : 0.7893 ± 0.0204
Recall    : 0.2997 ± 0.0175
F1 Score  : 0.4342 ± 0.0200
Confusion matrix saved to results/confusion_matrix.png

--- Character Decoding ---
Character Accuracy: 67.00% (67/100)
  reps=3 | ITR       : 24.40 bits/minute  (N=36, T=6.3s, reps=3)
Character Accuracy: 79.00% (79/100)
  reps=5 | ITR       : 19.15 bits/minute  (N=36, T=10.5s, reps=5)
Character Accuracy: 86.00% (86/100)
  reps=7 | ITR       : 15.79 bits/minute  (N=36, T=14.7s, reps=7)
Character Accuracy: 92.00% (92/100)
  reps=10 | ITR       : 12.45 bits/minute  (N=36, T=21.0s, reps=10)
Character Accuracy: 92.00% (92/100)
  reps=15 | ITR       : 8.30 bits/minute  (N=36, T=31.5s, reps=15)
```



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EEGNET:

Subject A:

```
--- Character Decoding ---
Character Accuracy: 46.00% (46/100)
  reps=3 | ITR       : 13.38 bits/minute  (N=36, T=6.3s, reps=3)
Character Accuracy: 50.00% (50/100)
  reps=5 | ITR       : 9.17 bits/minute  (N=36, T=10.5s, reps=5)
Character Accuracy: 67.00% (67/100)
  reps=7 | ITR       : 10.46 bits/minute  (N=36, T=14.7s, reps=7)
Character Accuracy: 77.00% (77/100)
  reps=10 | ITR       : 9.18 bits/minute  (N=36, T=21.0s, reps=10)
Character Accuracy: 89.00% (89/100)
  reps=15 | ITR       : 7.82 bits/minute  (N=36, T=31.5s, reps=15)
```

Subject B:

```
--- Character Decoding ---
Character Accuracy: 61.00% (61/100)
  reps=3 | ITR       : 21.00 bits/minute  (N=36, T=6.3s, reps=3)
Character Accuracy: 80.00% (80/100)
  reps=5 | ITR       : 19.56 bits/minute  (N=36, T=10.5s, reps=5)
Character Accuracy: 86.00% (86/100)
  reps=7 | ITR       : 15.79 bits/minute  (N=36, T=14.7s, reps=7)
Character Accuracy: 93.00% (93/100)
  reps=10 | ITR       : 12.70 bits/minute  (N=36, T=21.0s, reps=10)
Character Accuracy: 97.00% (97/100)
  reps=15 | ITR       : 9.18 bits/minute  (N=36, T=31.5s, reps=15)
```

ANALYSIS

LDA + Xdawn:

Subject A peaked at 91% accuracy (15 reps, 8.14 bits/min) with best ITR of 10.37 bits/min at 5 reps. Low recall (0.15) explains why A needs more repetitions. Subject B was stronger, 90% at 15 reps and best ITR of 26.19 bits/min at just 3 reps, with much better recall (0.46). CV accuracy was 83.8% for A and 87.4% for B.

SVM + Xdawn:

Subject A: 84% at 15 reps, best ITR 9.41 at 3 reps. Subject B: 92% at 15 reps, best ITR 24.40 at 3 reps. SVM showed perfect precision (1.0) for Subject A but near-zero recall (0.03), meaning it almost never predicted P300, yet still decoded characters reasonably through score averaging.

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Overall SVM underperformed LDA, confirming that after Xdawn the feature space is already linear and RBF kernel adds no benefit.

EEGNet:

Subject A: 89% at 15 reps, best ITR 13.38 at 3 reps. Subject B: 97% at 15 reps — highest accuracy across all models with best ITR 21.00 at 3 reps. Training loss steadily decreased across all 10 epochs for both subjects, indicating stable learning. EEGNet trades slightly lower ITR at low reps compared to LDA but wins on peak accuracy.

Overall: LDA achieves the best ITR (26.19 bits/min, Subject B), EEGNet achieves the best accuracy (97%, Subject B). Subject B consistently outperforms Subject A across all models, likely due to stronger and more consistent P300 amplitude.